Submission B solution to Problem 4

Theorem (fiber-filling and coarea reduction)

Let

$$R = \prod_{i=1}^{4}[0, R_i], \qquad S = \prod_{i=1}^{4}[0, S_i],$$

with $0 < R_1 \le R_2 \le R_3 \le R_4$ and $0 < S_1 \le S_2 \le S_3 \le S_4$. Orient S by

$$dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4.$$

Write

$$Q = [0, S_1] \times [0, S_2], \qquad B = [0, S_3] \times [0, S_4],$$

so $S = Q \times B$, and orient Q and B by $dx_1 \wedge dx_2$ and $dx_3 \wedge dx_4$, respectively. Let

$$\pi : S \to B, \qquad \pi(x_1, x_2, x_3, x_4) = (x_3, x_4),$$

and suppose

$$f : (R, \partial R) \to (S, \partial S)$$

is a Lipschitz finite piecewise smooth map of degree 1 with $\mathrm{Dil}_2(f) \le 1$. Put

$$F = \pi \circ f = (f_3, f_4), \qquad G = (f_1, f_2).$$

For $y \in B$, set

$$P_y = Q \times \{y\}.$$

Use the slicing convention normalized by

$$\langle [\![Q \times B]\!], \pi, y\rangle = [\![Q \times \{y\}]\!].$$

Then for a.e. $y \in \operatorname{int} B$, the slice

$$z_y = \langle [\![R]\!], F, y\rangle$$

is a relative integral 2-cycle in $(R, \partial R)$, and

$$f_{\#} z_y = [\![P_y]\!].$$

Consequently,

$$\mathrm{FillVol}_R(z_y) \geq \mathrm{FillVol}_S(P_y) = S_1 S_2 \ \mathrm{dist}(y, \partial B).$$

Moreover,

$$\int_B \mathrm{Area}(z_y)\, dy \leq \mathrm{Vol}(R).$$

Finally, let

$$B_0 = \left[\frac{S_3}{3}, \frac{2S_3}{3}\right] \times \left[\frac{S_4}{3}, \frac{2S_4}{3}\right].$$

At points where dF has rank 2, let

$$\sigma_1(x) \geq \sigma_2(x) > 0$$

be the two nonzero singular values of dF_x. Define the measure

$$d\mu = \frac{\sigma_2}{\sigma_1}\, dx$$

on the rank-two set of dF, and $d\mu = 0$ where $\operatorname{rank} dF < 2$. Then

$$\mu(F^{-1}(B_0)) \geq \frac{1}{9} S_1 S_2 S_3 S_4$$

and

$$\int_{B_0} \mathrm{Area}(z_y)\, dy = \int_{F^{-1}(B_0)} \sigma_1^2\, d\mu.$$

Thus, if in addition one has an estimate

$$\int_{F^{-1}(B_0)} \sigma_1^2\, d\mu \geq c_d \sqrt{\frac{S_3}{S_2}}\, \mu(F^{-1}(B_0))$$

for some constant $c_d > 0$, then

$$R_1 R_2 R_3 R_4 \geq \frac{c_d}{9} S_1 S_2^{1/2} S_3^{3/2} S_4.$$

Proof

Write $\mathbf{M}(T)$ for the mass of an integral current T, and write $[\![X]\!]$ for the integral current associated to an oriented Lipschitz manifold or rectangle X.

For a compact rectangle X, if $Z \in \mathbf{I}_m(X)$ satisfies $\operatorname{spt}\partial Z \subset \partial X$, define its relative filling volume by

$$\operatorname{FillVol}_X(Z) = \inf\left\{\mathbf{M}(Y) : Y \in \mathbf{I}_{m+1}(X),\ \operatorname{spt}(\partial Y - Z) \subset \partial X\right\}.$$

We use the standard slicing theorem for integral currents: if $T \in \mathbf{I}_m$ and Φ is Lipschitz into $\mathbb{R}^k$, then for a.e. a, the slice $\langle T, \Phi, a\rangle \in \mathbf{I}_{m-k}$ exists and

$$\partial\langle T, \Phi, a\rangle = \pm\langle \partial T, \Phi, a\rangle.$$

We also use slicing naturality: for Lipschitz H,

$$H_\#\langle T, \Phi \circ H, a\rangle = \langle H_\# T, \Phi, a\rangle$$

for a.e. a. Finally, we use the rectifiable-current area inequality

$$\mathbf{M}(H_\# T) \le \int J_k(dH|_{\operatorname{Tan} T})\, d\|T\|$$

for $T \in \mathbf{I}_k$.

First prove the fundamental current identity

$$f_\#[\![R]\!] = [\![S]\!].$$

Let

$$T = f_\#[\![R]\!].$$

Since f is Lipschitz, $T \in \mathbf{I}_4(S)$, and

$$\partial T = f_\# \partial [\![R]\!]$$

is supported in $f(\partial R) \subset \partial S$. Hence, in $\operatorname{int} S$,

$$\partial(T \llcorner \operatorname{int} S) = 0.$$

By the constancy theorem on the connected open set $\operatorname{int} S$,

$$T \llcorner \operatorname{int} S = m\, [\![\operatorname{int} S]\!]$$

for some $m \in \mathbb{Z}$. Therefore $T - m[\![S]\!]$ is supported in ∂S. Since ∂S has $\mathcal{H}^4$-measure zero, no nonzero integral 4-current can be supported there, so

$$T = m[\![S]\!].$$

Because f has degree 1, T represents the relative fundamental class $[S, \partial S]$, hence $m = 1$. Thus

$$f_{\#}[\![R]\!] = [\![S]\!].$$

Now apply slicing to $F = \pi \circ f$. For a.e. $y \in \operatorname{int} B$,

$$z_y = \langle [\![R]\!], F, y \rangle$$

is an integral 2-current, and

$$\partial z_y = \pm \langle \partial [\![R]\!], F, y \rangle,$$

so ∂z_y is supported in ∂R. Hence z_y is a relative integral 2-cycle in $(R, \partial R)$.

By slicing naturality,

$$f_{\#} z_y = f_{\#} \langle [\![R]\!], F, y \rangle = \langle f_{\#}[\![R]\!], \pi, y \rangle = \langle [\![S]\!], \pi, y \rangle.$$

With the stated product orientation and slice convention,

$$\langle [\![S]\!], \pi, y \rangle = [\![Q \times \{y\}]\!] = [\![P_y]\!].$$

Thus

$$f_{\#} z_y = [\![P_y]\!].$$

Next, $\mathrm{Dil}_2(f) \leq 1$ implies $\mathrm{Dil}_3(f) \leq 1$. Indeed, at a differentiability point of f, let

$$a_1 \geq a_2 \geq a_3 \geq a_4 \geq 0$$

be the singular values of df. The hypothesis gives $a_1 a_2 \leq 1$. Hence $a_2 \leq 1$, so $a_3 \leq 1$, and therefore

$$a_1 a_2 a_3 \leq a_1 a_2 \leq 1.$$

Thus $\|\Lambda^3 df\| \leq 1$ a.e.

Let $Y \in \mathbf{I}_3(R)$ be a relative filling of z_y, so

$$\operatorname{spt}(\partial Y - z_y) \subset \partial R.$$

Then

$$\partial(f_{\#} Y) - f_{\#} z_y = f_{\#}(\partial Y - z_y)$$

is supported in $f(\partial R) \subset \partial S$. Since $f_\# z_y = [\![P_y]\!]$, the current $f_\# Y$ is a relative filling of P_y in S. Also,

$$\mathbf{M}(f_\# Y) \le \int \|\Lambda^3 df\| \, d\|Y\| \le \mathbf{M}(Y).$$

Taking the infimum over all such Y gives

$$\mathrm{FillVol}_R(z_y) \ge \mathrm{FillVol}_S(P_y).$$

It remains to compute $\mathrm{FillVol}_S(P_y)$. Let

$$\delta(y) = \mathrm{dist}(y, \partial B).$$

For the upper bound, choose a shortest segment $\gamma \subset B$ from ∂B to y, oriented toward y. Then

$$C = [\![Q]\!] \times [\![\gamma]\!]$$

is a relative filling of P_y in S, and

$$\mathbf{M}(C) = S_1 S_2 \, \delta(y).$$

Hence

$$\mathrm{FillVol}_S(P_y) \le S_1 S_2 \, \delta(y).$$

For the lower bound, define

$$\delta(q) = \mathrm{dist}(q, \partial B) = \min\{q_3, S_3 - q_3, q_4, S_4 - q_4\}.$$

For $\eta > 0$, set

$$m_\eta(q) = -\eta \log\left(e^{-q_3/\eta} + e^{-(S_3-q_3)/\eta} + e^{-q_4/\eta} + e^{-(S_4-q_4)/\eta}\right).$$

Then

$$\delta(q) - \eta \log 4 \le m_\eta(q) \le \delta(q), \qquad |\nabla m_\eta(q)| \le 1.$$

Choose a smooth function $\chi_\eta : \mathbb{R} \to \mathbb{R}$ such that

$$\chi_\eta(t) = 0 \quad \text{for } t \le 0, \qquad 0 \le \chi_\eta'(t) \le 1, \qquad \chi_\eta(t) \ge t - \eta \quad \text{for } t \ge 0.$$

Set

$$\rho_\eta(q) = \chi_\eta(m_\eta(q)).$$

Then ρ_η is smooth,

$$\rho_\eta = 0 \quad \text{on } \partial B, \qquad |d\rho_\eta| \leq 1,$$

and for every fixed $y \in \operatorname{int} B$,

$$\rho_\eta(y) \to \delta(y) \qquad \text{as } \eta \to 0.$$

Define the smooth 2-form

$$\alpha_\eta = \rho_\eta(x_3, x_4)\, dx_1 \wedge dx_2.$$

It restricts to zero on ∂S: on the x_1- and x_2-faces because $dx_1 \wedge dx_2$ vanishes there, and on the x_3- and x_4-faces because $\rho_\eta = 0$ on ∂B.

Let $Y \in \mathbf{I}_3(S)$ be any relative filling of P_y, so

$$\operatorname{spt}(\partial Y - [\![P_y]\!]) \subset \partial S.$$

Since α_η vanishes on ∂S,

$$\partial Y(\alpha_\eta) = [\![P_y]\!](\alpha_\eta) = \rho_\eta(y) S_1 S_2.$$

By Stokes' theorem for currents,

$$Y(d\alpha_\eta) = \rho_\eta(y) S_1 S_2.$$

But

$$d\alpha_\eta = d\rho_\eta \wedge dx_1 \wedge dx_2,$$

whose comass is at most $|d\rho_\eta| \leq 1$. Therefore

$$\mathbf{M}(Y) \geq \rho_\eta(y) S_1 S_2.$$

Letting $\eta \to 0$ gives

$$\mathbf{M}(Y) \geq S_1 S_2\, \delta(y).$$

Taking the infimum over Y,

$$\operatorname{FillVol}_S(P_y) \geq S_1 S_2\, \delta(y).$$

Thus

$$\operatorname{FillVol}_S(P_y) = S_1 S_2 \ \operatorname{dist}(y, \partial B).$$

In particular, for $y \in B_0$,

$$\operatorname{dist}(y, \partial B) \geq \frac{S_3}{3},$$

because $S_4 \geq S_3$. Hence

$$\operatorname{FillVol}_R(z_y) \geq \frac{1}{3} S_1 S_2 S_3.$$

Now apply the coarea formula to the Lipschitz map $F : R \to B$. For every nonnegative Borel function g,

$$\int_B \int_{F^{-1}(y)} g \, d\mathcal{H}^2 \, dy = \int_R g \, J_2 F \, dx,$$

where

$$J_2 F = \|\Lambda^2 dF\|.$$

For a.e. y, the mass of the slice z_y is $\operatorname{Area}(z_y)$, so with $g = 1$,

$$\int_B \operatorname{Area}(z_y) \, dy = \int_R J_2 F \, dx.$$

Since $F = \pi \circ f$ and π is an orthogonal projection,

$$J_2 F \leq \|\Lambda^2 df\| \leq 1.$$

Therefore

$$\int_B \operatorname{Area}(z_y) \, dy \leq \operatorname{Vol}(R).$$

It remains to prove the two estimates involving μ. Fix a regular value $y \in B_0$, and let $x \in z_y$ be a point where f is differentiable and dF_x has rank 2. Then

$$T_x z_y = \ker dF_x.$$

Let

$$N_x = (T_x z_y)^\perp.$$

The restriction $dF_x|_{N_x} : N_x \to \mathbb{R}^2$ has singular values $\sigma_1(x) \geq \sigma_2(x) > 0$.

Choose a unit vector $n \in N_x$ such that

$$|dF_x(n)| = \sigma_1(x).$$

For any unit vector $v \in T_x z_y$, one has $dF_x(v) = 0$. Since

$$df_x = (dG_x, dF_x),$$

we get

$$df_x(v) = (dG_x(v), 0), \qquad df_x(n) = (dG_x(n), dF_x(n)).$$

In the orthogonal splitting $\mathbb{R}^4 = \mathbb{R}^2_Q \oplus \mathbb{R}^2_B$, the mixed $Q \wedge B$ component of

$$df_x(v) \wedge df_x(n)$$

is

$$dG_x(v) \wedge dF_x(n),$$

whose norm is

$$|dG_x(v)|\,|dF_x(n)| = |dG_x(v)|\,\sigma_1(x).$$

This mixed component has norm at most the norm of the full wedge. Therefore, using $\mathrm{Dil}_2(f) \le 1$,

$$|dG_x(v)|\,\sigma_1(x) \le |df_x(v) \wedge df_x(n)| \le 1.$$

Hence

$$|dG_x(v)| \le \sigma_1(x)^{-1}.$$

Taking the supremum over unit $v \in T_x z_y$,

$$\|dG_x|_{T_x z_y}\| \le \sigma_1(x)^{-1}.$$

Since $T_x z_y$ is two-dimensional,

$$J_2(G|_{T_x z_y}) \le \sigma_1(x)^{-2}.$$

Let $p_Q : S \to Q$ be the projection onto the first two coordinates. Since $G = p_Q \circ f$, for a.e. $y \in B_0$,

$$G_\# z_y = (p_Q)_\# f_\# z_y = (p_Q)_\# [\![P_y]\!] = [\![Q]\!].$$

Therefore

$$S_1 S_2 = \mathbf{M}([\![Q]\!]) = \mathbf{M}(G_\# z_y) \le \int_{z_y} J_2(G|_{T z_y})\, dA \le \int_{z_y} \sigma_1^{-2}\, dA.$$

Integrating over B_0,

$$\int_{B_0}\int_{z_y} \sigma_1^{-2}\, dA\, dy \geq S_1 S_2\, |B_0| = \frac{1}{9} S_1 S_2 S_3 S_4.$$

Now define

$$\widetilde{g}(x) = \sigma_1(x)^{-2}\, \mathbf{1}_{F^{-1}(B_0)\cap\{\operatorname{rank} dF=2\}}(x).$$

For $M > 0$, set $g_M = \min\{\widetilde{g}, M\}$. By the coarea formula,

$$\int_{B_0}\int_{z_y} g_M\, dA\, dy = \int_{F^{-1}(B_0)} g_M\, J_2F\, dx.$$

Letting $M \to \infty$ and using monotone convergence,

$$\int_{B_0}\int_{z_y} \sigma_1^{-2}\, dA\, dy = \int_{F^{-1}(B_0)\cap\{\operatorname{rank} dF=2\}} \sigma_1^{-2} J_2F\, dx.$$

On the rank-two set,

$$J_2F = \sigma_1\sigma_2.$$

Thus

$$\sigma_1^{-2} J_2F = \frac{\sigma_2}{\sigma_1}.$$

Hence

$$\int_{B_0}\int_{z_y} \sigma_1^{-2}\, dA\, dy = \int_{F^{-1}(B_0)} \frac{\sigma_2}{\sigma_1}\, dx = \mu(F^{-1}(B_0)).$$

Combining this with the previous inequality gives

$$\mu(F^{-1}(B_0)) \geq \frac{1}{9} S_1 S_2 S_3 S_4.$$

Finally, applying the ordinary coarea formula over B_0,

$$\int_{B_0} \operatorname{Area}(z_y)\, dy = \int_{F^{-1}(B_0)} J_2F\, dx.$$

On the rank-two set,

$$J_2F = \sigma_1\sigma_2 = \sigma_1^2 \frac{\sigma_2}{\sigma_1},$$

while both sides vanish on the rank < 2 set. Therefore

$$\int_{B_0} \operatorname{Area}(z_y)\, dy = \int_{F^{-1}(B_0)} \sigma_1^2\, d\mu.$$

Assume now that

$$\int_{F^{-1}(B_0)} \sigma_1^2 \, d\mu \geq c_d \sqrt{\frac{S_3}{S_2}} \, \mu(F^{-1}(B_0)).$$

Then

$$\int_{B_0} \mathrm{Area}(z_y) \, dy \geq c_d \sqrt{\frac{S_3}{S_2}} \, \mu(F^{-1}(B_0)) \geq \frac{c_d}{9} S_1 S_2 S_3 S_4 \sqrt{\frac{S_3}{S_2}}.$$

Since

$$\mathrm{Vol}(R) \geq \int_B \mathrm{Area}(z_y) \, dy \geq \int_{B_0} \mathrm{Area}(z_y) \, dy,$$

we obtain

$$R_1 R_2 R_3 R_4 = \mathrm{Vol}(R) \geq \frac{c_d}{9} S_1 S_2^{1/2} S_3^{3/2} S_4.$$

This completes the proof.